Project Summary

Overview

Ice-sheet science increasingly combines numerical models, ensemble data assimilation, observational archives, and scalable computing, but researchers still spend substantial effort installing software, translating data, rewriting scheduler workflows, and reconstructing computational context. This three-year CSSI Elements project will advance CryoStack from a promising developer-operated stack into secure, reusable, and measurable cyberinfrastructure for reproducible cryosphere computing. Existing code establishes feasibility: password/GitHub/ORCID identity, role-based administration and a Control Center, persistent workspaces and experiment histories, a modularized CryoLauncher interface, ICESEE data assimilation, LIVIST radar/borehole temperature exploration, a connector/relay path to Slurm and (for CryoLauncher's ISSM workflow on AWS) to an institutional MATLAB license, a manifest-driven Frozen Legacies historical-radar catalog, and a bring-your-own-account AWS path on which CryoLauncher's ISSM and Icepack workflows and a single-process ICESEE Lorenz-96 example have run end to end. It is not yet a finished community service: extension contracts, production-safe remote execution, scientifically qualified cloud and HPC runs, archival provenance, operations, governance, and adoption evidence remain incomplete.

The project has three objectives. O1 will deliver versioned application/model/dataset extension kits, automated CryoBench qualification, a production PISM coupling, and a second Frozen Legacies collection. O2 will harden an authentication-bound, allow-listed execution plane for local systems, Georgia Tech PACE, and project- or user-managed AWS accounts, with administrator-controlled computing allocations. O3 will create durable experiment/provenance services, a connected historical-data-to-model workflow, measurable operations, user evaluation, training, and contributor governance. Each extension will reuse shared identity, validation, execution, provenance, and deployment services.

Intellectual Merit

CryoStack will enable reproducible experiments in which common observation bundles and assimilation configurations can be evaluated across PISM, ISSM, Icepack, and idealized models, supporting studies of ice-sheet state, uncertain basal and rheological parameters, and structural uncertainty. The CI innovation is a clean, versioned separation of scientific and dataset adapters, execution backends, and gateway/operations services, validated by PISM and a second historical-data collection, neither of which shaped the interfaces. CryoBench will gate every extension on schema, numerical, scientific, security, and data-quality tests, including cross-backend scientific equivalence: whether a workflow remains scientifically consistent, within stated tolerances, across local, institutional-HPC, and cloud execution. A hardened, allow-listed control plane will replace the current broad-shell connector for securely bridging the gateway to protected institutional resources, and every run will produce a versioned experiment/result/provenance record spanning configuration, execution, backend, and application-specific diagnostics. PISM and one additional historical collection are the committed new integrations; the same contracts create a bounded path for future community models, data, and analysis tools without rebuilding the surrounding infrastructure.

Broader Impacts

Browser-first, containerized workflows will lower infrastructure barriers for students and researchers without dedicated research-software support. Three annual workshops, reusable tutorials, office hours, and paired contribution reviews will train participants in data assimilation, reproducibility, HPC, and scientific software engineering, prioritizing early-career researchers and under-

resourced institutions. Open releases and archival DOIs, accessible documentation, transparent governance, privacy-preserving metrics, and a five-year operations plan will establish a credible path from working prototypes to community-maintained cyberinfrastructure. The primary alignment is CISE/OAC; the domain alignment is GEO/OPP.